Experiment No. 09

Aim

To prepare a Risk Mitigation, Monitoring, and Management Plan (RMMM) for the Online Safari Boat Booking System.

1. Risk Identification in Online Safari Boat Booking System

| Risk ID | Date | Probability (%) | Risk Level | Impact | Risk Description | Current Status |
|---|---|---|---|---|---|---|
| R1 | 10-10-2026 | 45% | Medium | High | Unauthorized access to customer information | Open |
| R2 | 10-10-2026 | 60% | High | High | Double booking of safari boats for the same time slot | Open |
| R3 | 10-10-2026 | 40% | Medium | High | Incorrect booking amount or payment calculation | Open |
| R4 | 10-10-2026 | 35% | Medium | High | Server downtime or system crash during booking | Open |
| R5 | 10-10-2026 | 50% | Medium | Medium | Delay in displaying available boats and booking details | Open |
| R6 | 10-10-2026 | 25% | Low | Medium | Users facing difficulty using the booking system | Open |
| R7 | 10-10-2026 | 20% | Low | High | Loss of booking records during database updates | Open |

*Note: The probabilities are estimated values for this academic practical, not measured production statistics.*

2. Risk Mitigation Plan

| Risk ID | Mitigation Strategy |
|---|---|
| R1 | Use secure login, password hashing, role-based access control, and data encryption. |
| R2 | Validate boat availability before confirming a booking and prevent duplicate bookings using database transactions and constraints. |
| R3 | Validate booking charges, use automatic fare calculations, and test payment calculations thoroughly. |
| R4 | Perform regular database backups, monitor server health, and implement error handling and recovery mechanisms. |
| R5 | Optimize database queries, reduce unnecessary requests, and improve page-loading performance. |
| R6 | Provide a simple user interface, clear instructions, booking confirmation messages, and user support. |
| R7 | Schedule regular database backups, maintain transaction logs, and test database recovery procedures. |

3. Risk Monitoring Plan

Security Risks (R1): Review access logs, monitor suspicious login attempts, and check access permissions regularly.

Booking Risks (R2): Test boat availability checks and monitor duplicate booking attempts.

Payment Risks (R3): Verify fare calculations and review booking and payment records.

Technical Risks (R4, R7): Monitor server uptime, database health, backup completion, and recovery procedures.

Performance Risks (R5): Monitor page-loading time, database response time, and booking request processing.

Usability Risks (R6): Collect user feedback and identify difficulties faced during registration and booking.

4. Risk Management Plan

| Risk ID | Action if Risk Occurs (Contingency Plan) |
|---|---|
| R1 | Block unauthorized access, reset compromised credentials, investigate access logs, and notify affected users when necessary. |
| R2 | Stop duplicate confirmations, verify boat availability and booking records, and confirm the correct booking with the customer. |
| R3 | Recalculate the booking amount, correct the transaction record, and process any required refund or payment adjustment. |
| R4 | Restore the service, recover booking data from the latest valid backup if necessary, and notify users about service interruptions. |
| R5 | Investigate slow queries, optimize the database, and temporarily limit non-essential operations if required. |
| R6 | Provide user assistance, clarify booking steps, and improve confusing interface elements. |
| R7 | Restore the database from the latest valid backup, verify booking records, and reapply missing transactions where possible. |

Conclusion

The RMMM plan identifies the major risks associated with the Online Safari Boat Booking System and provides suitable mitigation, monitoring, and contingency strategies. It helps protect customer information, prevent duplicate bookings, maintain accurate payment records, and improve the reliability and availability of the system.